

RAMDULAR YADAV

ML Engineer | GCP Certified

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EDUCATION

Tribhuvan University — Purwanchal Campus

2021 – 2025

Bachelor of Computer Engineering

Equivalent GPA: 3.2/4.0 | First Division (65.87%)

Thesis: AI-Based Online Exam Proctoring System

WORK EXPERIENCE

Machine Learning Engineer (Full-Time) | Fusemachines

Sep 2025 – Present

- Built an AI-powered platform that converts e-commerce product reviews into structured insights and sentiment summaries.
- Developed a system to analyze customer feedback and generate product improvement and design recommendations.
- Created automated workflows to classify reviews and highlight key issues, trends, and customer needs.
- Produced visual and text-based outputs to support better product decision-making.
- Designed and deployed a demand forecasting pipeline to optimize inventory planning for a furniture retail brand, reducing stockouts and overstock risk.
- Built and compared multiple time series and ML models (Linear Regression, XGBoost, Prophet), selecting the best-performing model based on MAE and RMSE.
- Engineered features including seasonality trends, lag variables, and rolling statistics to improve forecast accuracy.
- Implemented the Prophet model for seasonality-aware forecasting and benchmarked it against baseline ML approaches.

AI/ML Intern | Fusemachines

May 2025 – Sep 2025

- Built an LLM-powered Interview Agent using LangChain for dynamic and human-like interview simulations.
- Designed and curated technical learning materials for the AI Fellowship program covering evaluation metrics, model stability, and self-supervised learning.

AI Fellowship | Fusemachines

Apr 2024 – Nov 2024

- Completed a competitive 6-month fellowship focused on Machine Learning, Deep Learning, Computer Vision, NLP, and MLOps.
- Developed and deployed end-to-end AI solutions including predictive models, web scraping pipelines, and real-time applications using Streamlit and Docker.
- Explored advanced concepts such as Transformers, Large Language Models, and Generative AI.
- Collaborated on team-based research projects, improving problem-solving, communication, and production-level AI system development.

Data Analyst | Karma Technology

Nov 2023 – May 2024

- Created and curated high-quality datasets crucial to the team's data-driven projects, contributing to effective decision-making.

RESEARCH PROJECTS

AI-Based Online Exam Proctoring System

Apr 2024 – Mar 2025

- Developed a real-time AI-powered proctoring system; fine-tuned YOLOv8 to detect banned objects from webcam input.
- Used OpenCV and MediaPipe to track face, eye gaze, and head direction; implemented audio monitoring for unusual sounds.
- Detected tab switching and full-screen exits; generated automated behavioural reports every 5 minutes.
- Built with Django backend, achieving 90% accuracy in detecting cheating behaviours.

Interview Agent

Jun 2025 – Present

- Built a multi-agent mock interview system using LangChain and Gemini LLM to simulate intelligent, job-specific interviews.
- Integrated `ResumeParserTool` and `JDParserTool` as LangChain tools, allowing LLMs to dynamically reason over parsed resumes and job descriptions.
- Designed agents for Q&A, Evaluation, Follow-Up, and Chitchat to create personalized, human-like interview experiences.

RAG-Based Advanced Multimodal Research Assistant

Jun – Jul 2025

- Developed an intelligent Q&A system for multi-format documents (PDF, Word, TXT) and images (charts, scanned notes).
- Enabled context-aware retrieval using ChromaDB for persistent vector storage of text and image embeddings.
- Implemented image captioning with Gemini Vision, semantic text chunking, and modular architecture for scalability.
- Built Streamlit UI and integrated RAGAS for response evaluation and hallucination reduction.

Fake News Detection Using NLP and Deep Learning

Feb – Jul 2025

- Developed a classification pipeline using ML, DL (LSTM, CNN), and transformers on WELFake and FakeNewsNet datasets.
- Achieved 98.2% accuracy on WELFake using fine-tuned BERT; 73.5% generalization accuracy using a BERT + CNN + LSTM hybrid on FakeNewsNet.
- Deployed a real-time prediction interface with Streamlit and containerized the solution using Docker.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, SQL, Bash, HTML/CSS
- **AI & Agentic Frameworks:** LangChain, RAG Systems, Prompt Engineering, Generative AI
- **Web & API Development:** FastAPI, Flask, Django, RESTful APIs, Streamlit
- **Machine Learning & Data Science:** scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Time Series Forecasting
- **Deep Learning & Computer Vision:** TensorFlow, Keras, PyTorch, OpenCV
- **Natural Language Processing:** HuggingFace Transformers, SpaCy, NLTK, BERT, GPT, NER, Summarization
- **Databases:** MySQL, PostgreSQL
- **Cloud & Deployment:** Google Cloud Platform (GCP), Docker, Model Deployment
- **Workflow Automation:** n8n, Claude Code
- **Soft Skills:** Leadership, Teamwork, Communication, Analytical Thinking, Problem Solving

CERTIFICATIONS

- Google Cloud Certified Professional Machine Learning Engineer (GCP)
- Microdegree in Artificial Intelligence — Fusemachines AI Fellowship
- Introduction to Machine Learning
- Intro to ChatGPT and Generative AI